University of Technical Education HCMC
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SOFTWARE DEVELOPMENT ENVIRONMENTS

Chapter 02

Unified Modeling Language - UML

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Lesson 02

System behavior



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Objectives

After the lesson, you will be able to:

- Define system behavior
- Define use cases and actors
- Understand how to document use cases
- Use a use case diagram to show the actors, the use cases, and their interactions
- Define scenarios for use cases



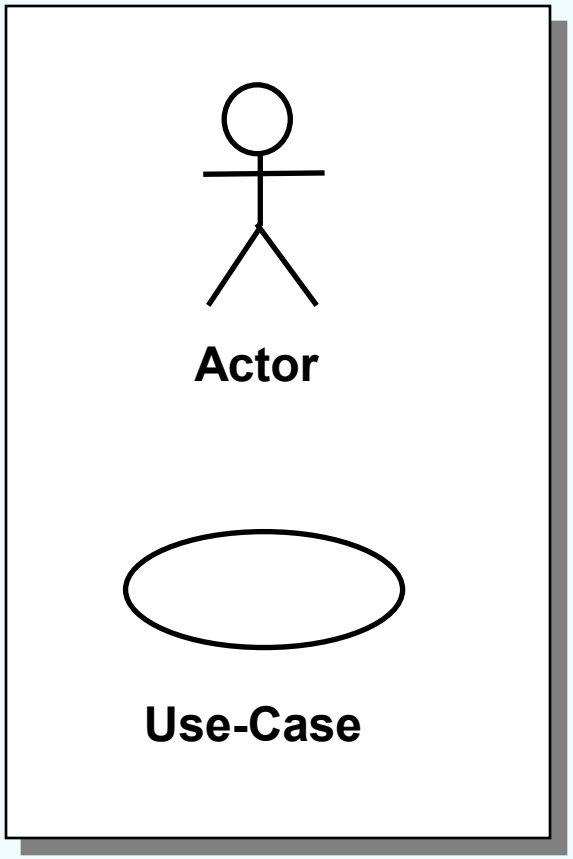
What is System Behavior?

- **System behavior is how a system acts and reacts**
  - The outwardly visible and testable activity of a system
- **System behavior is captured in use cases**
  - They describe the system, its environment, and the relationship between the system and its environment



Major Concepts in Use-Case Modeling

- An actor represents anything that interacts with the system
- A use case is a sequence of actions a system performs that yields an observable result of value to a particular actor



What is a Use-Case Model ?

- A use-case model is a model of the system's intended functions (use cases) and its surroundings (actors)
- The same use-case model is used in requirements analysis, design, and test

The use case model's primary purpose is to communicate the system's functionality and behavior to the customer or end user



Benefits of a Use-Case Model

- **The use case model**
  - Is used to communicate with the end users and domain experts
    - Provides buy-in at an early stage of system development
    - Insures a mutual understanding of the requirements



Benefits of a Use-Case Model

- **The use case model**
  - Is used to identify
    - Who will interact with the system and what the system should do
    - What interfaces the system should have
  - Is used to verify
    - All requirements are captured
    - That the developers have understood the requirements



Actors

- Actors are not part of the system, they can represent roles a user of the system can play
- An actor may actively interchange information with the system
- An actor may be a passive recipient of information
- An actor can represent a human, a machine or another system

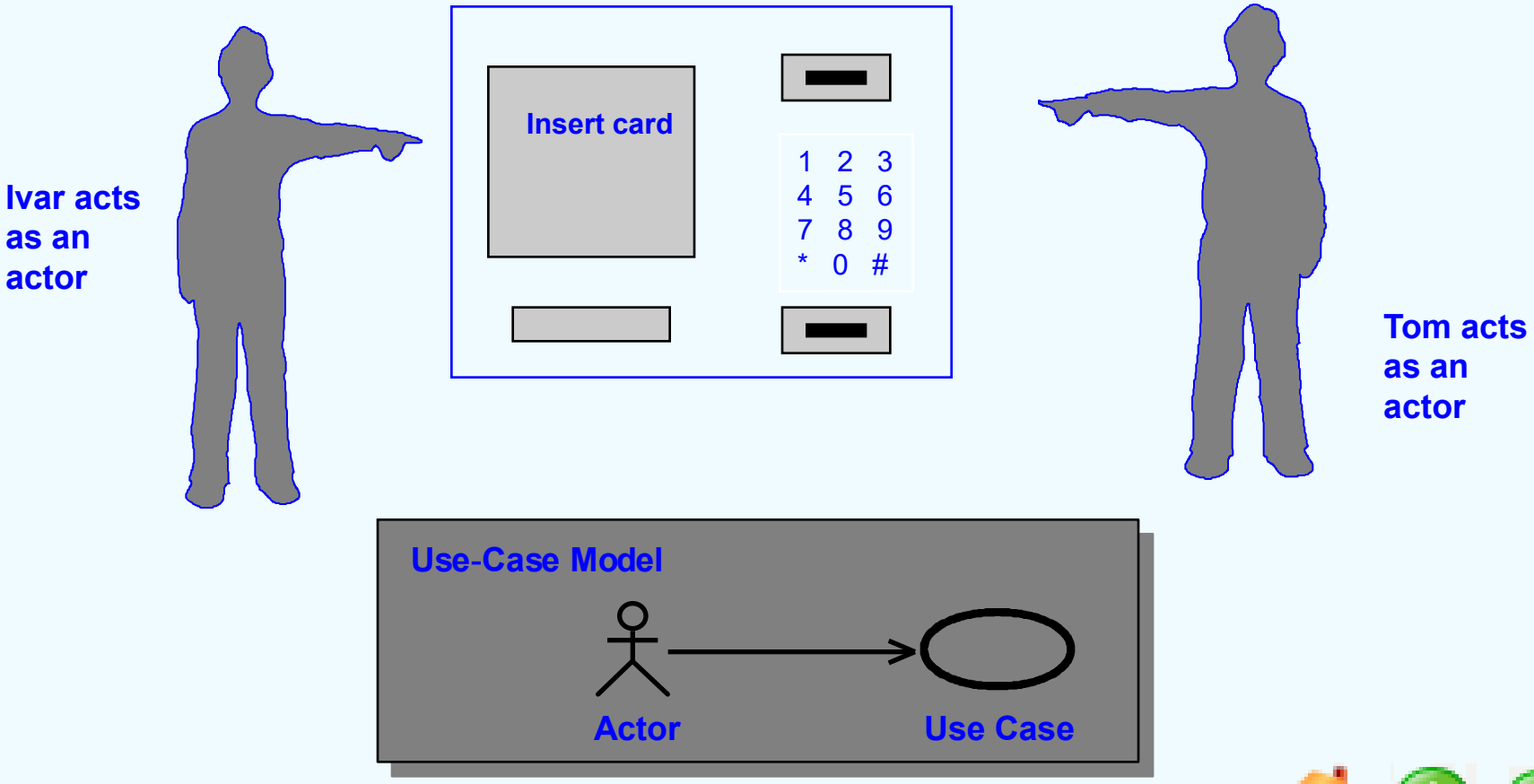


Finding Actors: Useful Questions

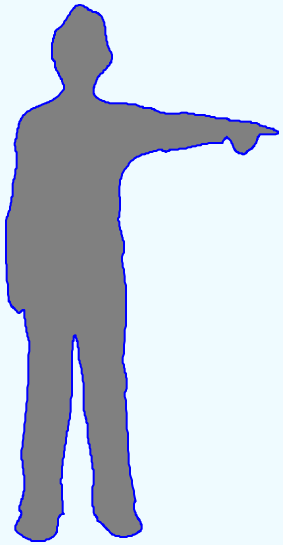
- Who is interested in a certain requirement?
- Where in the organization is the system used?
- Who will supply the system with this information, use this information, remove this information?
- Who will use this function?
- Who will support and maintain the system?
- Does the system use an external resource?
- What actors do the use cases need?
- Does one actor play several different roles? Do several actors play the same role?



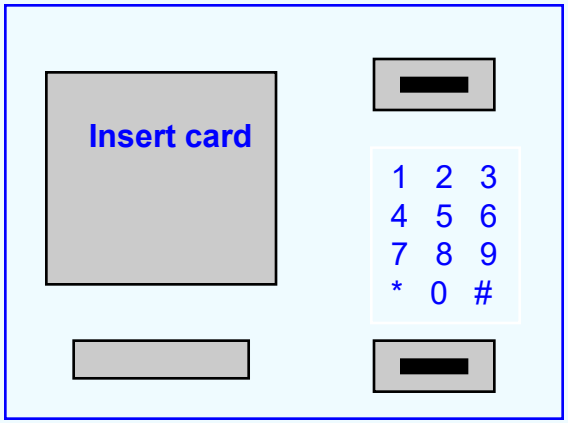
Instances of Actors



A User May Act as Several Actors

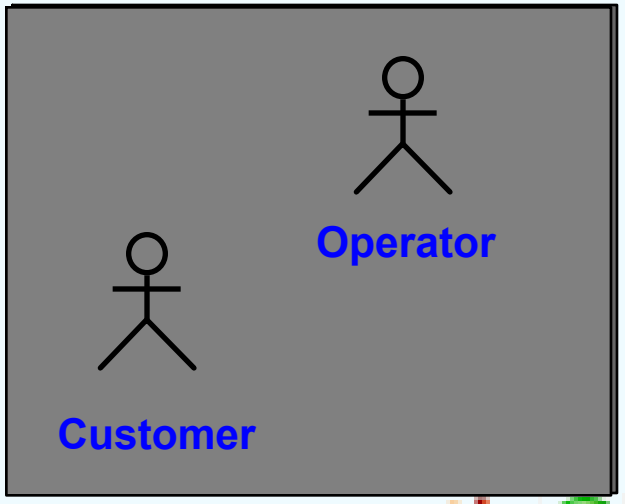


Charlie

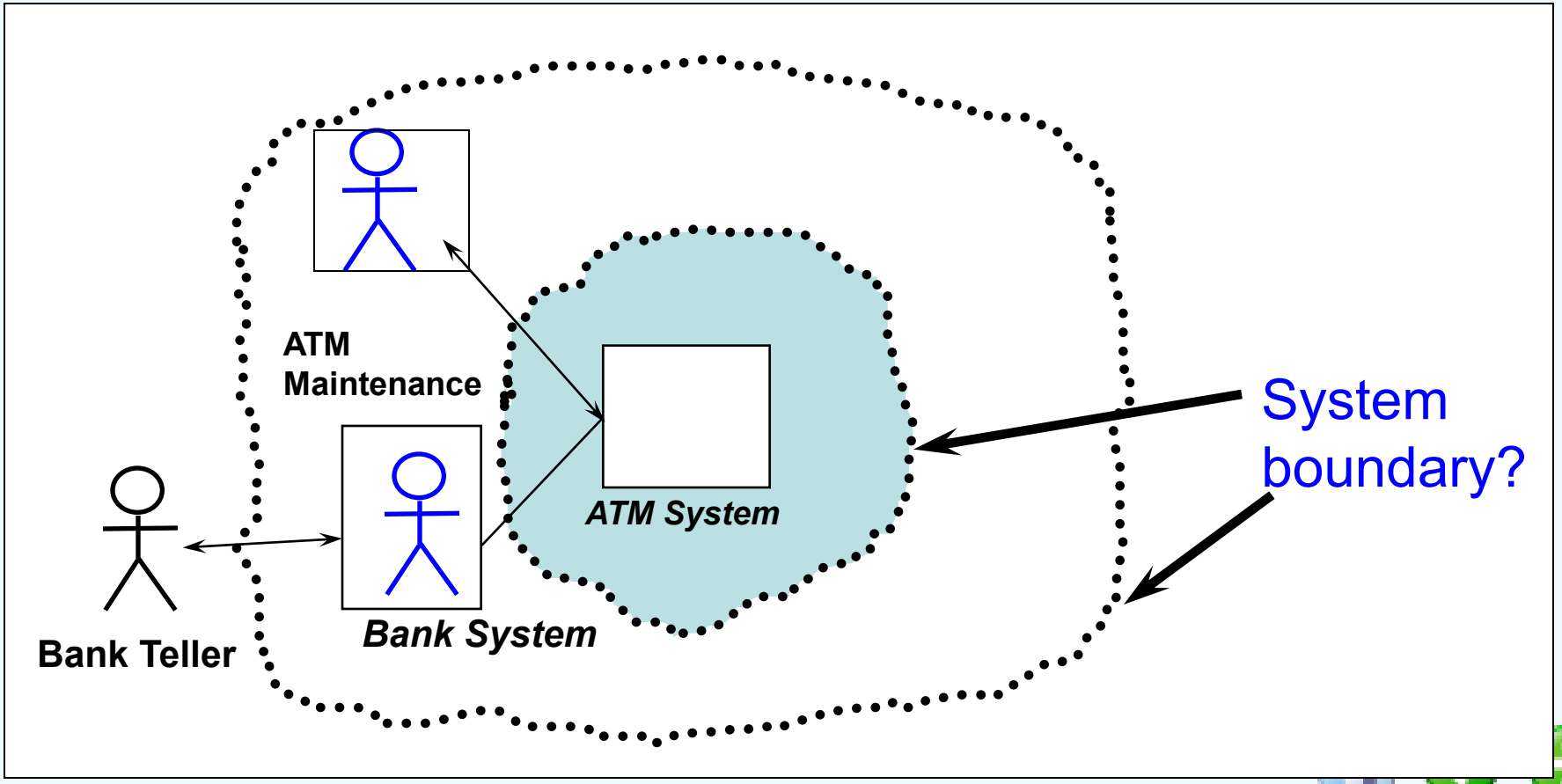


Charlie as operator

Charlie as customer

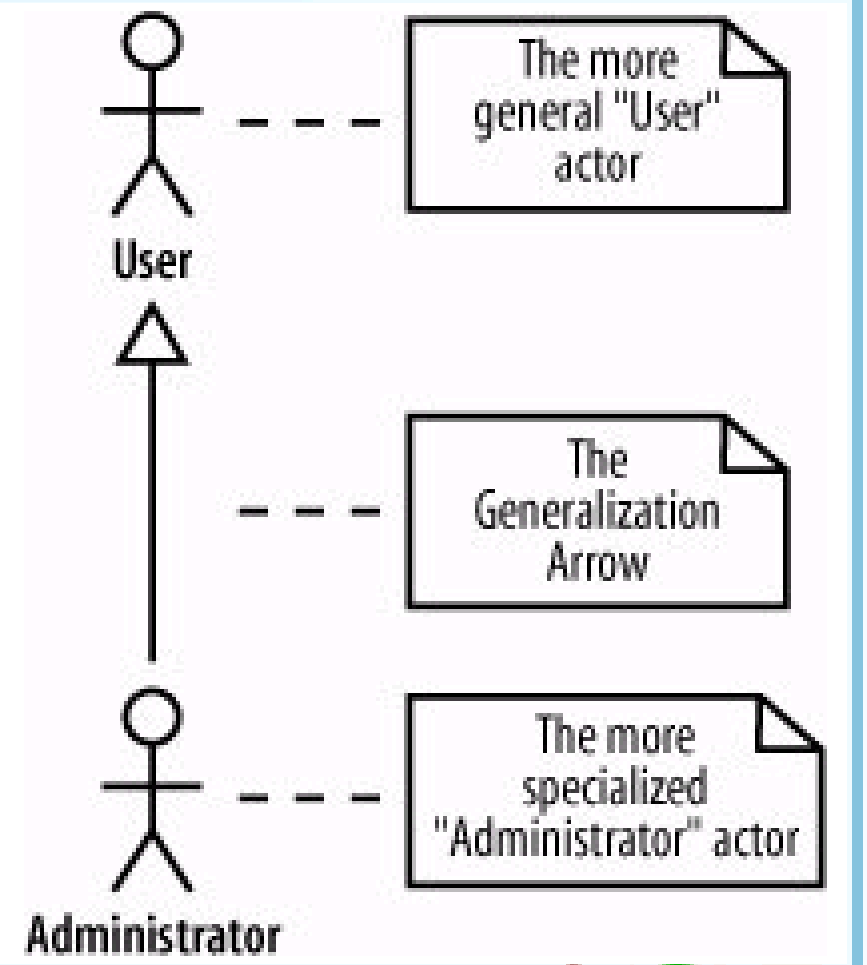


Actors and System Boundaries

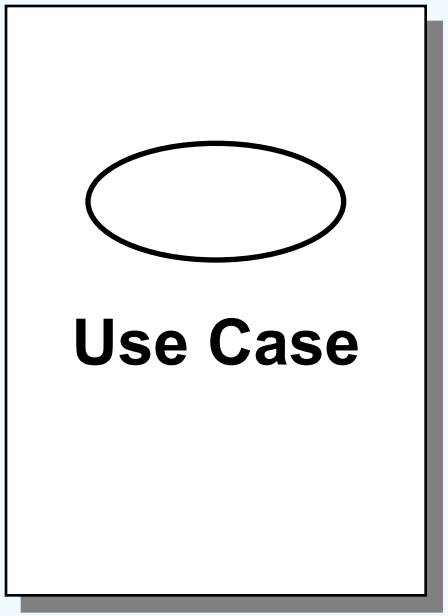


Relationships between actors

- Different actors may have some relationship to each other



Use Cases



- A use case models a dialogue between actors and the system
- A use case is initiated by an actor to invoke a certain functionality in the system
- A use case is a complete and meaningful flow of events
- Taken together, all use cases constitute all possible ways of using the system

Finding Use Cases: Useful Questions

- What are the tasks of this actor?
- Will the actor create, store, change, remove or read information in the system?
- What use case will create, store, change, remove, or read, this information?
- Will the actor need to inform the system about sudden, external changes?
- Does the actor need to be informed about certain occurrences in the system?
- Does the system supply the business with the correct behavior?
- What use cases will support and maintain the system?
- Can all functional requirements be performed by the use cases?



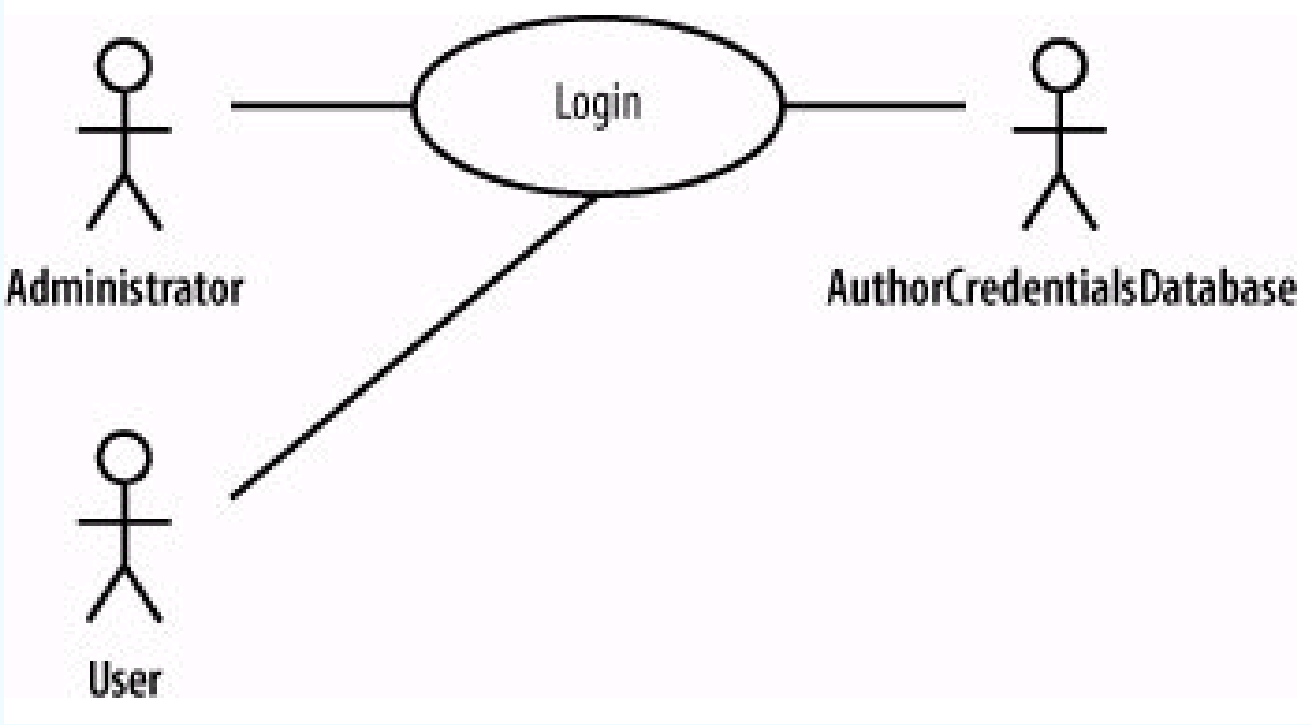
Sources of Information for Use Cases

- System specifications/problem statement
- Domain relevant literature
- Interviews with domain experts
- Personal knowledge of the domain
- Legacy systems



The Use Case Diagram

- A use case diagram is drawn to illustrate that use cases and actors interact by sending stimuli to one another



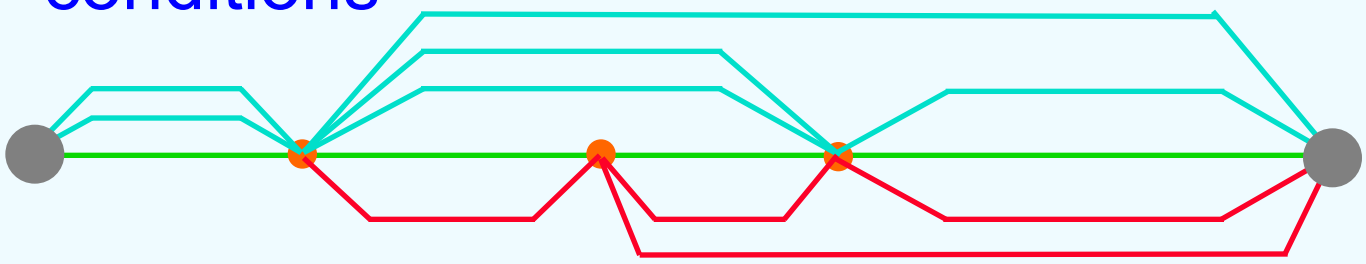
Use Case Documentation

- Use cases are documented in
  - A brief description
    - **The purpose of the use case in a few lines**
  - Detailed flow of events
    - **Description of the primary and alternate flow of events that occur when the use case is initiated**
  - Documentation should read like a dialogue between the actor and the use case
- Both documents are written in terms the customer will understand



Use Case Flow of Events

- Each use case:
  - Has one normal, basic, sequence of transactions
  - May have several alternative sequences of transactions
  - Usually has several exceptional sequences of transactions handling erroneous situations
  - May also have well defined pre- and post-conditions



Use Case Flow of Events (2)

- Describe only the events that belong to the use case, and not what happens in other use cases
- Avoid vague terminology such as "for example", "etc." and "information"



Use Case Flow of Events (3)

- The flow of events should describe
  - How and when the use case starts and ends
  - When the use case interacts with the actors
  - What information is exchanged between an actor and the use case
    - **Do not describe the details of the user interface**
  - The basic flow of events
  - Any alternate flow of events



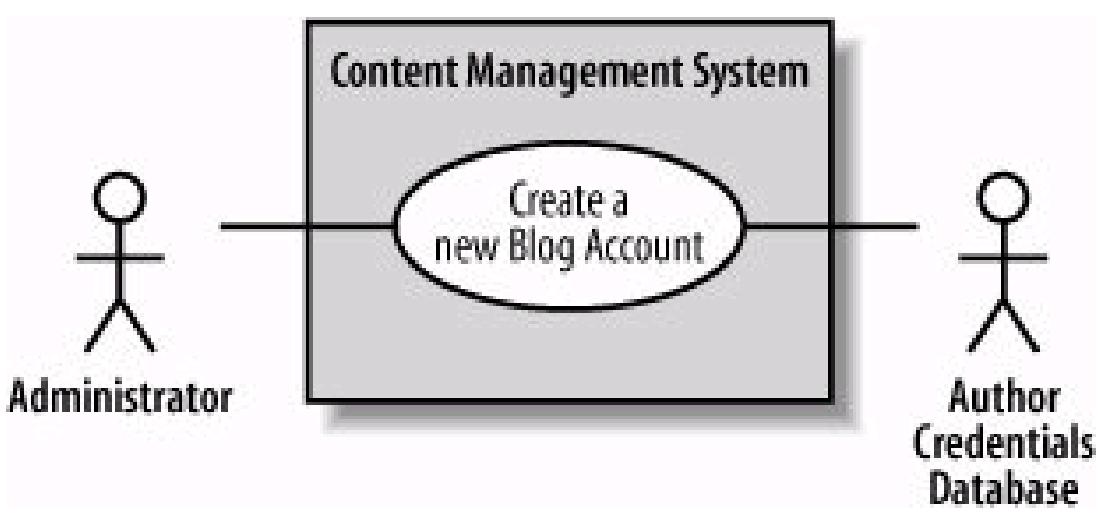
Who Reads Use-Case Documentation ?



- Customers -- approve what the system should do
- Users -- gain system understanding
- System Developers -- document system behavior
- Reviewers --examine the flow of events
- System Analysts (System Designer) -- provide the basis for analysis and design
- System Tester -- used as a base for test cases
- Project Leader -- provide input to project planning
- Technical Writer -- basis for writing the user's guide

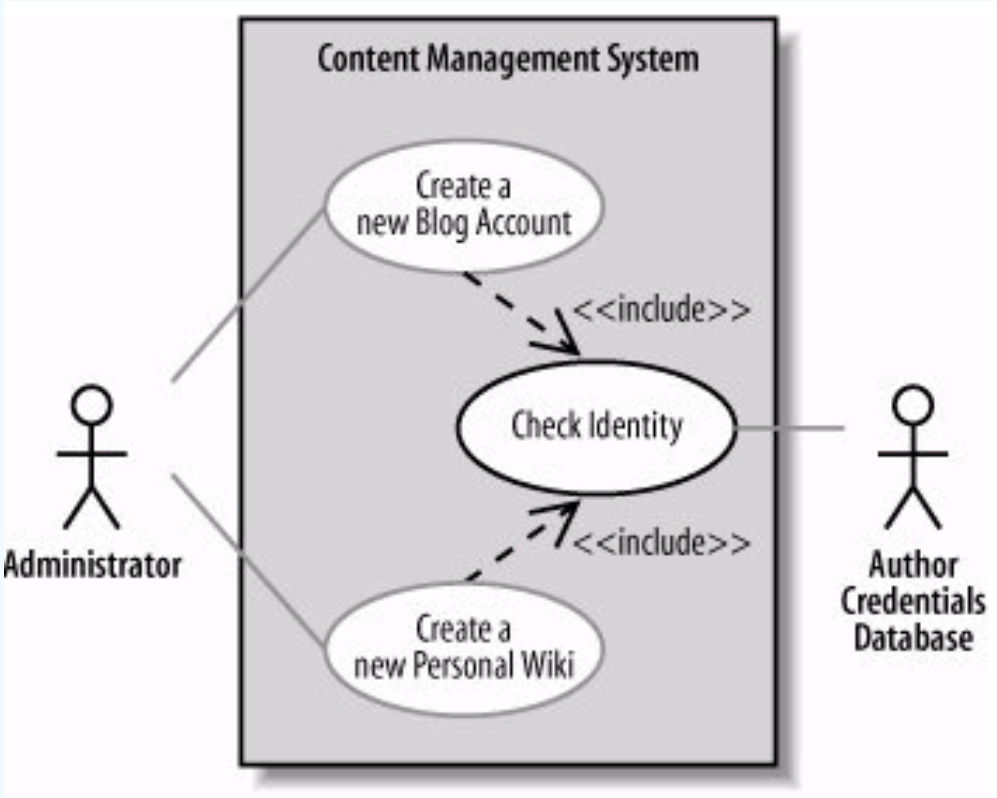
Use case description sync

- **Author Credetials Database help binding every description with its use case**



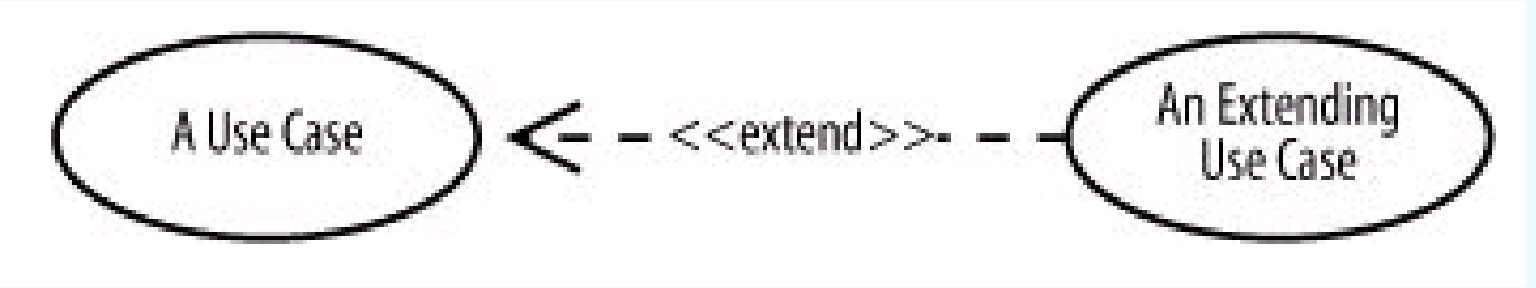
Use case relationships

- **<<include>> relationship supports reuse between use cases**



Use case relationships (2)

- **<<extend>> relationship is somehow similar to <<include>>, but the reuse is optional**



Course Registration Example

– At the beginning of each semester students may request a course catalogue containing a list of course offerings for the semester. Information about each course, such as professor, department, and prerequisites will be included to help students make informed decisions.



Course Registration Example (2)

– The new system will allow students to select four course offerings for the coming semester. In addition, each student will indicate two alternative choices in case the student cannot be assigned to a primary selection. Course offerings will have a maximum of ten students and a minimum of three students. A course offering with fewer than three students will be canceled. Once the registration process is completed for a student, the registration system sends information to the billing system so the student can be billed for the semester.



Course Registration Example (3)

- Professors must be able to access the on-line system to indicate which courses they will be teaching. They will also need to see which students signed up for their course offerings.
- For each semester, there is a period of time that students can change their schedule. Students must be able to access the system during this time to add or drop courses.



Brief Description

Register for Courses Use Case

1.1 Brief Description

This use case is initiated by a student. It provides the capability for the student to create, delete, modify and/or review a course schedule for a given semester.



Flow of Events
Register for Courses Use Case

2.1 Pre-conditions

None

2.2 Main Flow

This use case begins when the student enters the student id number. The system verifies that the student id number is valid (E-1) and prompts the student to select the current semester or a future semester (E-2). The student enters the desired semester. The system prompts the student to select the desired activity: CREATE, REVIEW, MODIFY, PRINT, DELETE, or QUIT.



Flow of Events
Register for Courses Use Case (2)

If the activity selected is

CREATE, the A-1: Create a New Schedule subflow is performed.

REVIEW, the A-2: Review a Schedule subflow is performed.

MODIFY, the A-3: Modify a Schedule subflow is performed.

PRINT, the A-4: Print a Schedule subflow is performed .

DELETE, the A-5: Delete a Schedule subflow is performed.

QUIT, the use case ends.



Flow of Events
Register for Courses Use Case (3)

2.3 Alternate Flows

A-1: Create a New Schedule

The system displays a blank schedule screen. The student enters 4 primary course offering numbers and 2 alternate course offering numbers (E-3). The student then submits the request for courses. For each primary course selected the system will check that prerequisites are satisfied (E-4) and add the student to the course offering if the course offering is open (E-5). The system prints the student schedule (E-6) and sends billing information to the billing system for processing (E-7). The use case then begins again.



Flow of Events
Register for Courses Use Case (3)

A-2: Review a Schedule

The system retrieves (E-8) displays the following information for all all course offerings for which the student is registered: course name, course number, course offering number, days of the week, time, location, and number of credit hours. When the user indicates that s/he's through reviewing, the use case begins again.



Flow of Events
Register for Courses Use Case (4)

A-3: Modify a Schedule

The system checks that the final date for changes has not been exceeded (E-9). The system retrieves (E-8) and displays the following information for all all course offerings for which the student is registered: course name, course number, course offering number, days of the week, time, location, and number of credit hours. The system prompts the user to select the desired activity: DELETE A COURSE OFFERING, ADD A COURSE OFFERING, or QUIT.

If the activity selected is

DELETE A COURSE OFFERING, the A-6: Delete a Course Offering subflow is performed.

ADD A COURSE OFFERING, the A-7: Add a Course Offering subflow is performed.

QUIT, the system prints the student schedule (E-6) and the use case then begins again.



Flow of Events
Register for Courses Use Case (5)

A-4: Print a Schedule

The system prints the student schedule (E-6). The use case begins again.

A-5: Delete a Schedule

The system retrieves (E-8) and displays current schedule information. The system asks to user to confirm the delete of the schedule. If accepted, the schedule is removed from the system. If the delete is not confirmed, the operation is canceled and the use case begins again.

A-6: Delete a Course Offering

The student enters the number of the offering to delete. The system asks to user to confirm the delete of the course offering. If accepted, the course offering is removed from the student schedule. If the delete is not confirmed, the operation is canceled and the use case alternate flow begins again.



Flow of Events
Register for Courses Use Case (7)

A-7: Add a Course Offering

The student enters the course offering to add. The system will check that prerequisites are satisfied (E-4) and add the student to the course offering if the course offering is open (E-5). The use case alternate flow begins again.

E-1: Exception Flows

E-1: An invalid student id number is entered. The user can re-enter a student id number or terminate the use case.

E-2: An invalid semester is entered. The user can re-enter the semester or terminate the use case.

E-3: The entered course offering number is not valid (format). The user can re-enter a valid number or terminate the use case.

E-4: The user has not satisfied all pre-requisite requirements. The user is informed that a course can not be scheduled and why. If possible, an alternate course is substituted. The use case continues.



Flow of Events
Register for Courses Use Case (8)

E-5: The user is informed that a selected course offering is closed. If possible, an alternate course is substituted. The use case continues.

E-6: The schedule cannot be printed. The information is saved and the user is informed that a print schedule request must be re-submitted. The use case continues.

E-7: The system will save all billing information and re-submit it to the billing system at a later date. The use case continues

E-8: The system cannot retrieve schedule information. The use can then starts at the beginning.

E-9: The system informs the user that a schedule cannot be modified. The use case then starts at the beginning.



What Are Scenarios ?

- A scenario is an instance of a use case
  - It is one flow through a use case
- Each use case will have a web of scenarios
  - Primary scenarios (happy day scenarios)
    - **Normal flow - the way the system should work**
  - Secondary scenarios
    - **Exceptions to the primary scenario**



One Scenario for the "Register for Courses" Use Case

John enters the student ID number 369 52 3449 and the system validates the number. The system asks which semester. John indicates the current semester and chooses create a new schedule.

From a list of available courses, John selects the primary courses English 101, Geology 110, World History 200, and College Algebra 110. He then selects the alternate courses Music Theory 110 and Introduction to Java Programming 180.

The system determines that John has all the necessary prerequisites and adds him to the course rosters.

The system indicates that the activity is complete. The system prints the student schedule and sends billing information for four courses to the billing system for processing.



Secondary Scenarios

- What are some of the secondary scenarios for the “Register for Courses” use case ?



Secondary Scenarios (2)

- Some secondary scenarios to consider are:
  - Student does not select 4 primary courses
  - Primary course is not available
  - Primary and secondary courses are not available
  - Cannot add student to the course roster
  - Cannot create the student schedule



How Many Scenarios Are Needed?

- Simple answer: as many as one needs to understand the system being developed
- Rule of thumb:
  - Primary scenarios
    - **Elaborate approximately 80% of these scenarios**
  - Secondary scenarios
    - **Elaborate a few of the interesting and high-risk secondary scenarios**



Summary

- System behavior is how a system acts and reacts
- The behavior of a system may be characterized by a set of use cases
- A use case is some functionality performed by the system in response to a stimulus from an external actor
  - **They provide a vehicle for capturing the requirements about a system from a user point of view**



Summary (2)

- A actor is someone or something that must interface with the system under development
- A use case diagram is a graphical depiction of the system which shows the actors and use cases identified for the system
- Use case documentation consists of a brief description and a flow of events





Excercises

- Xét hệ thống quản lý thư viện với mô tả như sau:
  - Hệ thống có khả năng quản lý tất cả các đầu sách hiện có trong thư viện, nhập sách mới, thanh lý sách cũ, tra cứu sách…
  - Có khả năng quản lý độc giả, thêm, bớt độc giả, cấp thẻ độc giả…
  - Trước khi mượn sách, độc giả có thể sử dụng hệ thống để tra cứu tên, vị trí, tình trạng của những sách mình cần mượn cũng như số lượng sách còn được phép mượn



Excercises

- Sau khi đã vào thư viện lựa chọn những quyển sách cần mượn, độc giả sẽ chuyển sách cho thủ thư tại quầy để nhập thông tin cho mượn sách
- Khi độc giả mang sách đến trả, thủ thư nhận sách,cập nhật lại thông tin trả sách, tiến hành phạt nếu trả trễ, hoặc yêu cầu đền bù nếu làm mất sách.
- Người quản trị có khả năng quản lý và sử dụng toàn bộ hệ thống

– Hãy vẽ lược đồ use-case mô tả các chức năng của hệ thống thư viện nêu trên





THE END

